Handspell

Learn the ASL fingerspelling alphabet by spelling the real things around your desk.

Gaming track | New Experience | Target: Best New Gaming Experience

AT A GLANCE

Track / division	Gaming (educational casual game) / New Experience. Productivity is a possible alternative track.
Primary prize	Best New Gaming Experience ($100,000)
Fallback special award	Best Accessibility Forward Experience ($25,000), the most realistic win for this idea
Session length	The Daily 5 is 5 minutes; a full sitting is 10 minutes
Room requirement	A desk; room labels from Space Setup enrich the Room Hunt mode
Core tech	WebXR hand tracking, a hand-shape classifier, motion matching for J and Z, plane detection with semantic labels
Biggest risk	Letters where the thumb hides behind the fingers (A, E, M, N, S, T) are hard for hand tracking
Respect requirement	Consult Deaf signers; present it as fingerspelling practice, not as teaching ASL

The one-line pitch

A learning game for the American Sign Language (ASL) fingerspelling alphabet. Your hands are the whole controller and the whole curriculum: spell floating words to break them, spell the real objects in your room to claim them, and grow a garden of mastered letters on your desk.

SECTION 01

The idea

Fingerspelling is how signers spell names, places and words that have no dedicated sign, one handshape per letter. It is also the first thing most ASL learners study, and one of the hardest to practise alone: there is nobody to tell you whether your handshape is right, and you cannot see your own hand from the angle other people see it.

Handspell solves both problems. Hand tracking checks every handshape as you make it, and a mirrored 'coach view' shows your hand from the viewer's side, the way someone reading your fingerspelling would see it. Practice is wrapped in short, satisfying game modes played on your desk.

Why it stands out

It is the most hands-first idea imaginable: the hands are not only the input method but the subject being learned. It also has built-in visible progress (letters mastered, words per minute), which is exactly what the Best Reason to Come Back and Accessibility awards reward.

SECTION 02

Respect and responsibility

Sign languages belong to Deaf communities. A hearing developer building a sign-language product must be careful not to misrepresent the language or suggest that an app replaces learning from Deaf teachers.

- **Scope honestly:** call it fingerspelling practice. It teaches the 26 handshapes of the ASL manual alphabet, not ASL grammar or vocabulary.
- **Consult early:** ask one or more Deaf ASL users or teachers (paid, credited) to review handshapes, feedback wording and the video. Online Deaf-led communities and local Deaf organisations are the right places to ask.
- **Point outward:** the app ends every session with a pointer to learning from Deaf teachers and Deaf-led resources.
- **Avoid 'curing' language:** it is a language-learning game, not an accessibility aid for Deaf people and not a translator.

SECTION 03

Game modes

Mode	How it plays	Learning purpose
Letter Garden	Learn letters in small groups. Each letter you master plants a small glowing plant on your desk that grows as your accuracy improves.	Introduces handshapes; the garden shows progress.
Word Rain	Short words drift down toward the desk. Spell one to burst it before it lands.	Builds speed and letter-to-letter transitions.
Room Hunt	Labels appear on real objects around you (TABLE, WINDOW, DOOR, SCREEN). Spell one to claim it; it glows.	Connects spelling to real things; uses the room meaningfully.
Daily 5	A 5-minute session built by spaced repetition from your weakest letters and transitions.	Retention; a daily habit.
Name Tag	Spell your own name and the names of your friends and pets.	Personal, motivating, and the most common real-world use of fingerspelling.

SECTION 04

The first five minutes

Time	What happens
0:00	Passthrough. A small seed floats above the desk. A soft voice and caption: 'Show me your hand.' The player raises a hand.
0:20	A ghost hand appears making the letter L (index finger up, thumb out), a letter that is easy to track. The player copies it; the ghost fills with colour as the match improves, and the seed sprouts.
1:00	Two more easy letters, I (little finger up) and D (index up, other fingers touching the thumb). Each correct match plants a sprout.
2:00	The coach view appears: a mirrored hand shows how the letter looks to someone watching. The player sees their own L from the front for the first time.
3:00	First word, built only from letters already learned: a lid appears on the desk and the player spells L-I-D. The lid pops open and the next letters fly out.
4:30	The garden now has three sprouts on the desk. The session ends with 'Come back tomorrow and they will grow.'

Note: the first letters are chosen for recognisability, not alphabetical order. Letters with clear, unoccluded finger positions (B, L, Y, V, W, I, D, F) come first; letters that are hard for tracking come later with more forgiving checks.

SECTION 05

Why passthrough and the real room matter

This is the weakest part of the idea for the judges' 'purposeful passthrough' criterion, so it has been designed deliberately.

- **Room Hunt spells your real surroundings.** Space Setup gives semantic labels to surfaces and furniture (table, wall, window, door, couch, screen, and others). Handspell pins spelling targets to those real objects, so the vocabulary comes from your room.
- **Your real hand stays visible.** Learners need to see their actual fingers; passthrough keeps them visible alongside the virtual coach hand, rather than replacing them with an avatar hand.
- **The garden lives on your desk** and stays there between sessions (saved with a spatial anchor), so progress is physically present in your space.

> **Honest assessment**
>
> Even with these features, most of Handspell would work in full VR. Judges may mark it down under Experience Design for passthrough. That is the main reason it ranks below Tabletop Beat and Shadowplay.

SECTION 06

Recognition design

Static letters (24 of 26)

For each frame, the app turns the 25 tracked joints into features that do not depend on hand size or position: the bend of each finger joint, the spread between neighbouring fingers, the thumb tip's position relative to the index and middle knuckles, and the palm's orientation relative to the head. A small classifier (nearest-neighbour or a tiny neural network running in the browser) compares these features to recorded examples of each letter.

- **Training data:** record examples from the developer and several volunteers of different hand sizes, both hands, under normal lighting.
- **Smoothing:** a letter is accepted when it is the top match for about 300 ms with enough confidence.
- **Confusable groups:** A, S, T, M, N and E differ mostly in where the thumb sits, which the headset cameras often cannot see from behind the hand. Early levels accept these as a group; later levels use stricter checks only if device testing shows they are reliable.

Motion letters (J and Z)

J traces a hook with the little finger; Z traces a Z with the index finger. These are recognised by recording the fingertip's path and comparing it with a template path using dynamic time warping (a standard technique for matching gestures performed at different speeds).

Transitions

Fluent fingerspelling is about moving smoothly between letters. The app tracks accuracy for letter pairs (for example E to R), and spaced repetition schedules weak pairs as well as weak letters.

SECTION 07

Progression and reasons to come back

System	Description
Letter Garden	Each letter is a plant on the desk. It grows as your accuracy on that letter improves and wilts slightly if you do not practise it for a week.
Words per minute	A personal speed record for common words, shown as a line over weeks.
Spaced repetition	Letters and pairs move between review boxes (a Leitner system), so the Daily 5 always targets what you are about to forget.
Streak	Consecutive days of Daily 5 practice.
Room collection	Every object you have spelled in your room keeps a faint glow.

SECTION 08

Accessibility

- **Fully visual feedback.** No part of the game depends on sound, so Deaf and hard-of-hearing players can use it fully; all audio has captions.
- **Left-handed mode** mirrors every letter.
- **Adjustable strictness** so players with limited finger mobility can still progress.
- **Slow mode** for Word Rain.
- **High-contrast mode** for the coach hand and labels.

SECTION 09

Art direction and sound

Calm and bright: a soft garden palette of greens and warm whites on the real desk. Letters are shown as clean, friendly hand illustrations and a translucent 3D coach hand. Sound is optional and gentle: a soft chime on a correct letter, a rising tone on a completed word.

SECTION 10

Technical architecture (IWSDK)

Feature	Used for
Hand tracking	Joint poses for the classifier; the coach hand is built from the player's own joint data, mirrored.
Plane detection with semantic labels	Room Hunt targets and desk placement for the garden.
Anchors	Keeping the garden in the same place on the desk.
IndexedDB	Accuracy history, spaced repetition state, streaks.

System	Responsibility
HandFeatureSystem	Computes the feature vector from joints every frame, without allocating memory.
LetterClassifierSystem	Runs the classifier and smoothing; emits letter events with confidence.
MotionLetterSystem	Records fingertip paths and matches J and Z.
SpellingSessionSystem	Drives the current mode's word, timer and scoring.
CoachViewSystem	Mirrors the player's hand into the viewer's perspective.
GardenSystem	Grows or wilts plants from accuracy history.
RepetitionSystem	Schedules letters and pairs for review.

SECTION 11

Risks and mitigations

Risk	Severity	Mitigation
Occluded-thumb letters misclassified	High	Group confusable letters early; measure per-letter accuracy in week 1; be honest in the app about which letters are checked loosely.
Cultural misstep	High	Consult and credit Deaf signers; honest scoping; point to Deaf-led learning.
Passthrough judged as not purposeful	Medium	Room Hunt, garden on the desk, real hands visible. Accept some risk.
Classifier works for the developer's hands only	Medium	Collect examples from several people with different hand sizes.
Existing sign-language learning apps	Low	Differentiate on real-time checking, the coach view and room-based vocabulary.

Week-1 spike

1. Record 20 samples of each letter from 2 or 3 people.
2. Train a quick nearest-neighbour classifier and measure accuracy per letter on a held-out set.
3. Decide which letters are reliable, which need grouping, and whether the concept holds.

SECTION 12

Seven-week build plan

Week	Dates	Goal	Exit criterion
Week 1	29 Sep - 4 Oct	Recognition spike; contact Deaf consultants	Per-letter accuracy measured; consultant conversation scheduled
Week 2	5 - 11 Oct	Classifier in the headset; coach view; Letter Garden for 8 letters	Learn-and-check loop works on device
Week 3	12 - 18 Oct	All 26 letters including J and Z; Word Rain	Every letter can be practised
Week 4	19 - 25 Oct	Spaced repetition; Daily 5; progress storage	A return visit feels different from the first
Week 5	26 Oct - 1 Nov	Room Hunt; Name Tag; consultant review of handshapes and wording	Feedback incorporated
Week 6	2 - 8 Nov	Accessibility options; performance; playtests with learners	72 fps; testers improve across sessions
Week 7	9 - 15 Nov	Polish; record video; write submission	Video and text ready
Ship	16 - 18 Nov	Final checks; submit on 17 Nov	Submitted early

SECTION 13

Scope

Must ship	Nice to have	Cut
26 letters with honest grouping; coach view; Letter Garden; Word Rain; Daily 5; left-hand mode	Room Hunt; Name Tag; words-per-minute chart	Full ASL signs; other sign languages; two-handed alphabets such as BSL (which would need robust two-hand tracking)

SECTION 14

Demo video plan (under 3 minutes)

Time	Beat	Shot
0:00 - 0:12	Hook	A hand spells W-I-N-D-O-W and the real window across the room lights up.
0:12 - 0:45	The problem	Learning fingerspelling alone means no feedback. Show the coach view: your hand as others see it.
0:45 - 1:40	First five minutes	Real-time: the seed, the first letters, the garden sprouting.
1:40 - 2:20	Play	Word Rain and Room Hunt.
2:20 - 2:55	Growth	The garden after two weeks of practice; words-per-minute chart; credit to the Deaf consultants.

SECTION 15

Submission draft

Name	Handspell
Tagline (under 140 characters)	Practise the ASL fingerspelling alphabet with real-time hand checking, and grow a garden of mastered letters on your desk.
Track / division	Gaming / New Experience
Target launch	Web release Q1 2027 after further review with Deaf consultants

Description (first draft)

Fingerspelling is the first thing most ASL learners study and one of the hardest to practise alone. Nobody tells you whether your handshape is right, and you never see your hand the way others see it. Handspell fixes both.

Hand tracking checks every letter as you make it, and a mirrored coach view shows your hand from the viewer's side. Short game modes turn practice into play: burst falling words by spelling them, spell the real objects in your room to light them up, and grow a garden on your desk where every mastered letter is a plant.

We built Handspell with the Immersive Web SDK, using a classifier on hand-joint features for static letters and path matching for J and Z. Handshapes and wording were reviewed by Deaf ASL users, and every session points learners toward Deaf-led teaching.

SECTION 16

How it maps to the judging criteria

Criterion (25% each)	Evidence in Handspell
Innovation and Creativity	Hands as both controller and curriculum; coach view that no mirror can provide.
Experience Design	Short daily sessions, visible growth, no text walls. Passthrough use is the weak spot.
Technical Implementation	Custom hand-shape classifier and motion matching running in the browser.
Polish and Presentation	Calm, clear visuals; a strong social-good story.

Hard requirement and design guidelines

Test	What the rules ask	How this idea passes
Hands-first (hard requirement)	Complete the entire experience without ever pairing a controller.	Hands are the only input; menus use pinches.
Airplane seat test	Every interaction works within a two-foot radius, seated.	All interaction is at desk height in front of the player.
One bus stop test	Fast cold start, clean pause/resume, a satisfying moment in 10 minutes or less.	The Daily 5 is exactly 5 minutes; progress saves after every word.
Take it away test	Remove any third-party service and a project still stands.	No third-party services; the classifier runs locally.

Competition rules that shape this plan

- **One prize per entry.** Rule 7 limits each entry to a single prize, so this brief aims at one track prize and names one special award as a fallback rather than trying to stack awards.
- **Submission:** a public HTTPS link (GitHub Pages, Vercel or similar) that stays live until the winner announcement (about 11 Dec 2026), a public YouTube/Vimeo video under 3 minutes, and the Devpost form (name, 140-character tagline, track, division, description of about 500 words, hand-interaction notes, target launch date).
- **No changes after the deadline.** Judging uses the build as it stood at 18 Nov 2026, 12:00 PT. Plan to submit on 17 Nov and freeze the hosted build.
- **Content rules:** English (or English subtitles), no identifiable real people other than entrants, no brand names or logos, all music, art and fonts properly licensed or original.
- **Video rules:** real footage from a Quest or the XR simulator; do not lean on AI-generated video to carry the pitch.
- **Performance floor:** judges expect at least 60 fps on Quest hardware. Target 72 fps.

SECTION 17

Honest weaknesses

- Recognition limits on thumb-occluded letters are real and cannot be fully solved with current hand tracking.
- Passthrough is less essential than in the other ideas, which costs points in Experience Design.
- The Gaming track is crowded with pure games; Handspell may be seen as 'edutainment'.
- Getting cultural consultation right takes time and care, and must be started in week 1.

SECTION 18

Scorecard and verdict

Dimension	Score / 10	Comment
Originality	7	Sign-learning apps exist; real-time checking and the coach view are new.
Only possible in MR	4	Mostly works in VR too.
Works in untested rooms	10	Needs only a desk; Room Hunt is a bonus.
Solo feasibility in 7 weeks	7	Simple art; the classifier is the main work.
Prize ceiling	$100k / $25k	Track prize is a stretch; the Accessibility award is a realistic target.

Verdict

A meaningful project with a strong accessibility story, but it scores lower on the passthrough and originality criteria than the top two ideas. Choose it if the social impact matters most to you, and treat the Accessibility award as the main target.